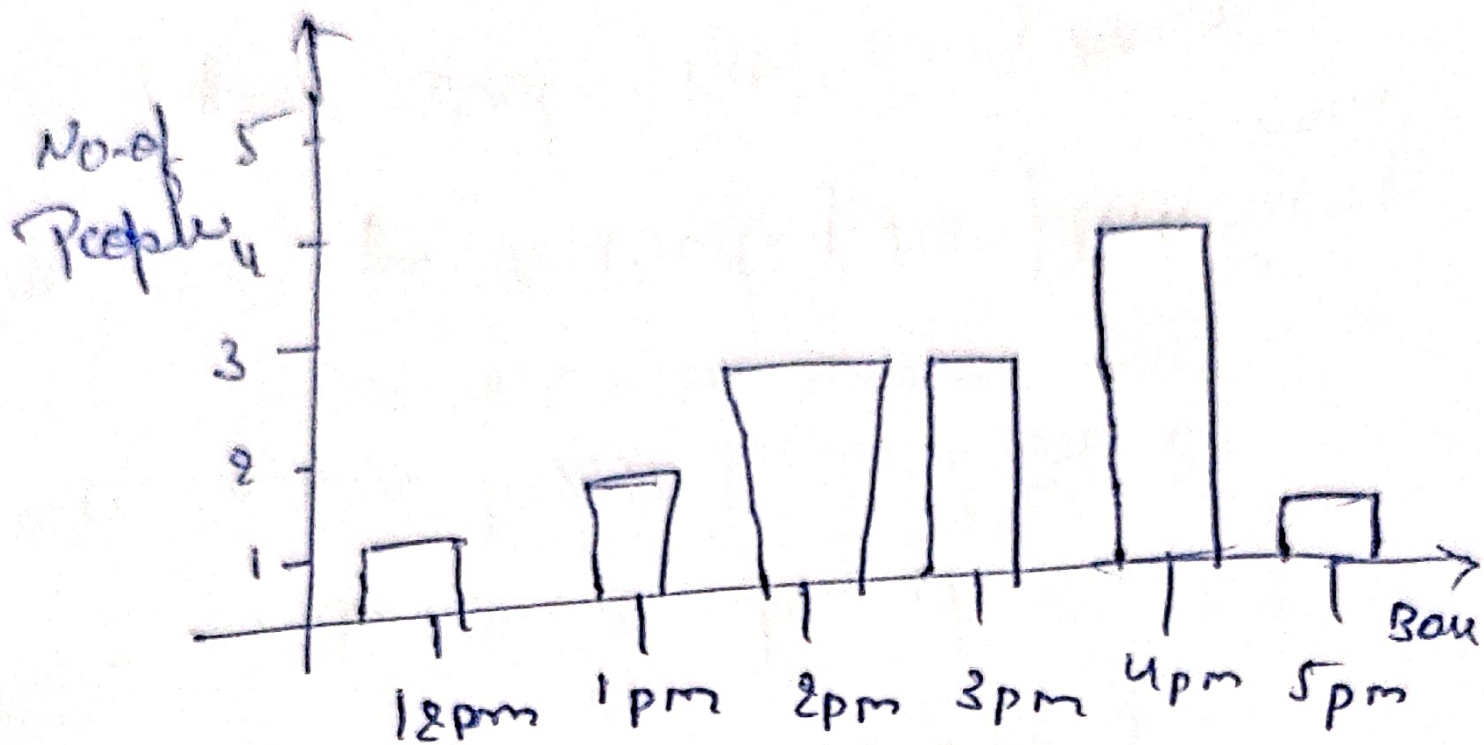


Poisson distributed

A discrete probability distribution that Express the probability of a given number of events occurring in a fixed interval of time if these events occur with a known constant mean rate and Independence of the time since the last event

- ① Discrete random variable (p.m.f)
- ② Describe the number of events occurring in a fixed time interval

Eg:- No. of people visiting hospital Every hour
No. of people visiting Bank Every hour.



$\lambda = 3 \Rightarrow$ Expected no. of events occurring at every time interval

PMF

$$P(X=5) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \boxed{\lambda = 3}$$

$$= \frac{e^{-3} 3^5}{5!} = 0.101 = 10.1\%$$

$$P(X=4) + P(X=5)$$

Mean

$$\text{Mean} = E(X) = \mu = \lambda * t$$

Variance 